S.S.V.P.S's B. S. Deore Polytechnic, Dhule

Department of Computer Engineering

Academic Year : 2026-2027

SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	SPI – 315003	Date :

Topic : Industrial Immersive Training Platform

Roll No.	Name of Student
14	Purva Vikas Bhamare
34	Shubhangi Sanjay Kedar
39	Prajakta Chetan Mahale
46	Mansi Pradip Nandre

Contents

- Certificate
- Abstract

1. Introduction
2. Literature Review
3. Information to Seminar topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
    - 6. 1 Description of problem statement.
    - 6. 2 Scope and objectives.
    - 6. 3 Stake holder
    - 6. 4 seminar exsicussion plan
    - 6.5 Platform/ Equipment/ Resources identification.

1. Bibliography
2. References

Abstract

- Immersive technology is becoming useful in education, industrial training, and technical skill development.
- Traditional training needs physical machines, trained instructors, and workshop facilities, so students get limited practical exposure.
- The Industrial Immersive Training Platform is a web-based application to learn industrial machines through interactive digital content.
- Features: machine search, machine information, interactive 3D visualization, safety instructions, simulated troubleshooting, quizzes, and progress tracking.
- Runs in a web browser, with no VR headset required. Intended for education only; it does not replace workshop training or official safety procedures.

1. Introduction

- Industrial training helps students understand machines, their components, working principles, and safety procedures.
- Access to real machines is limited by high cost, maintenance needs, and safety risks.
- VR, AR, and interactive 3D visualization let learners explore digital machine models.
- Proposed project: a web-based application with machine information, 3D models, component identification, simulated troubleshooting, safety quizzes, and progress tracking.
- Objective: make machine learning more accessible and interactive, while supporting (not replacing) workshop training.

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
1.	VR Training for KUKA Robot Operators (Pratticò & Lamberti, 2021)	VR, Robotics	Case Study	Train robot operators virtually	Limited real-world testing
2.	AR Tools for Aeronautic Maintenance (Loizeau et al., 2021)	AR	Field Evaluation	Support Maintenance Workers	Industry-specific
3.	Situated Visualization for Maintenance Training (BagheriFard et al., 2025)	AR, Visualization	User Study	Improve maintenance learning	Limited evaluation

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
4.	Immersive Technologies for Safety Training (Babalola et al., 2023)	VR, AR, XR	67 Articles	Review safety training methods	Varied study quality
5.	XR and Workers' Safety (Dodoo et al., 2025)	XR, VR, AR	Literature Review	Study XR in high-risk industries	Adoption challenges
6.	VR-ManuSafe (Fengjin et al., 2026)	VR, Unity	8 Injury Types	Teach manufacturing safety	Limited scope

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
7.	An Emerging Immersive Technology – A Survey (Pavithra et al., 2020)	VR, AR, MR	Literature Survey	Explore immersive technologies	General overview
8.	Immersive Technology: Uses and Challenges (Handa et al.)	VR, AR, MR	Literature Review	Discuss uses and challenges	Limited practical analysis
9.	State of Immersive Technology Research (Suh & Prophet, 2018)	VR, AR, MR	Research Literature	Analyze research trends	Depends on available studies

2. Literature Survey

Sr. No.	Aapproach Type	Technology Used	Dataset	Key Objective	Limitations
10.	Web-Based Digital Twin for E-Learning (Weis et al., 2025)	Web 3D, Digital Twin	7 Professionals	Support interactive learning	Small pilot study
11.	Immersive Learning for Machine Operators (Pletz & Zinn, 2020)	VR, Video Analysis	13 Participants	Evaluate operator training	Small sample size
12.	Industrial Machine Identification Using AR (Kavya et al., 2021)	Unity, Vuforia, ARCore	Prototype	Identify machines using AR	Platform limitations

3. Information to seminar topic

- **Industrial machines:** used in manufacturing, processing, assembly, and packaging; the platform gives each machine's name, use, components, and working principle.
- **Types:** Manufacturing (drilling, milling, lathe), Material Handling (conveyors, lifting), Assembly, Packaging, and Industrial Robots.
- **3D visualization:** rotate, zoom, and view a machine from different angles; selecting a component shows its name and basic information.
- **Component identification:** labelled 3D models and diagrams, e.g. a drilling machine has a drill bit, spindle, worktable, base, and column.



Figure 3.2: 3D Model of a CNC Lathe Machine with Major Components.



Figure 3.1: 3D Model of a CNC Milling Machine with Major Components.

4. Advantages / Disadvantages

Advantages

- Interactive learning
- Easy access through a browser, no VR headset
- Better understanding of machine structure
- Self-paced learning
- Safety awareness
- Simulated troubleshooting practice
- Progress tracking
- Less dependence on physical equipment

Disadvantages

- Cannot replace hands-on experience
- 3D models take skill and time to build
- Needs a reasonably capable device
- Needs a stable internet connection
- Limited simulation of real faults
- Content accuracy affects learning
- Needs regular updates
- Quizzes do not replace formal safety training

5. Conclusion

- The Industrial Immersive Training Platform is a proposed web-based application that supports industrial machine learning through interactive digital technology.
- It brings machine information, 3D visualization, component identification, simulated troubleshooting, and safety quizzes into one place.
- Users can explore machine models and learn about components in a web browser, without a VR headset.
- It cannot fully replace real equipment experience; its value depends on model quality, content accuracy, and successful implementation.
- Future scope: more machine models, more training activities, and improved interactive features.

6. Project Initiation

- **6.1** Description of problem statement
- **6.2** Scope and objectives
- **6.3** Stakeholders
- **6.4** Seminar discussion plan
- **6.5** Platform / equipment / resources identification

6.1 Description of problem statement

- Industrial training is important in technical education, but students may not get access to different machines because of high cost, limited availability, and maintenance needs.
- Learning only from textbooks and static images makes machine components and working principles hard to understand.
- **Need:** an interactive learning platform with machine information and digital visualization.
- **Solution:** a web-based environment to explore machines, identify components, practise simulated troubleshooting, and learn basic safety procedures.

6.2 Scope and objectives

Scope

- A web-based platform for basic industrial machine training with machine categories and search, 3D visualization, component identification, simulated troubleshooting, safety quizzes, user accounts, and progress tracking. For education only; supplements workshop training.

Objectives

- Develop a web-based industrial training platform
- Provide information on machines and their components
- Enable interactive 3D exploration of machine models
- Provide basic component identification activities
- Support learning through simulated troubleshooting
- Improve safety awareness through quizzes
- Let users review their learning progress

6.3 Stakeholders

- **Students:** learn about industrial machines and their components.
- **Teachers:** use the platform as a supplementary teaching resource.
- **Factory employees:** revise basic machine-related concepts.
- **Institution administrators:** manage institution users and learning resources.
- **Platform administrators:** manage user accounts, machine information, and content.
- **Project developers:** design, develop, test, and maintain the application.

6.4 Seminar discussion plan

- Introduction to industrial immersive technology
- Need for interactive industrial machine training
- Overview of the Industrial Immersive Training Platform
- Proposed system and its features
- Technologies required for development
- Advantages and disadvantages
- Possible applications and future improvements
- Conclusion and questions from the audience

6.5 Platform / Equipment / Resources

- **Laptop / Computer:** develop and test the application
- **Antigravity:** assist with application development and coding
- **HTML and CSS:** structure and style web pages
- **JavaScript:** application logic and interactivity
- **React:** build the user interface
- **Three.js:** display and interact with 3D machine models
- **MySQL Database:** store user and machine data
- **3D Machine Models:** digital representations of industrial equipment
- **Web Browser:** access and test the application
- **Internet Connection:** online tools, libraries, and services

Bibliography

- **Books**
  - Burdea, G. C., & Coiffet, P. (2003). *Virtual Reality Technology* (2nd ed.). Wiley-IEEE Press.
  - Craig, A. B. (2013). *Understanding Augmented Reality: Concepts and Applications*. Morgan Kaufmann.
  - Sherman, W. R., & Craig, A. B. (2003). *Understanding Virtual Reality: Interface, Application, and Design*. Morgan Kaufmann.
- **Websites**

https://doi.org/10.1016/j.ssci.2023.106214

https://doi.org/10.1016/j.ssci.2025.106804

https://doi.org/10.1016/j.simpa.2025.100790

References

- Pratticò, F. G., & Lamberti, F. (2021). *Towards the Adoption of Virtual Reality Training Systems for the Self-Tuition of Industrial Robot Operators: A Case Study at KUKA.*
- Loizeau, Q., et al. (2021). *Methodology for the Field Evaluation of the Impact of Augmented Reality Tools for Maintenance Workers in the Aeronautic Industry.*
- BagheriFard, Z., et al. (2025). *Situated Visualization Towards Manufacturing Maintenance Training: Scoping Review, Design and User Study.*
- Babalola, A., et al. (2023). *Applications of Immersive Technologies for Occupational Safety and Health Training and Education: A Systematic Review.*
- Dodoo, J. E., et al. (2025). *XR and Workers' Safety in High-Risk Industries: A Comprehensive Review.*
- Lei, F., Singh, D., & Chun, L. M. (2026). *VR-ManuSafe: Virtual Reality Learning Environment for Manufacturing Safety Training.*

Thank you